



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University

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College of Informatics and Computing Sciences

PROJECT PROPOSAL
OOP 101 – OBJECT-ORIENTED PROGRAMMING
Computer Science Department
1st Semester, AY 2026-2027

I. TEAM INFORMATION

Team Name:	L o a f code		
Team Number:	5	Section:	CS - 2102

TEAM MEMBERS			
No.	SR-Code	Name	GitHub Username
1	25-01345	Entienza, Ashenden Jairus V.	Harukii @Haru-Mb
2	25-03908	Hernandez, Diandre Daivik I.	Qlychn @Qlychn
3	25-09477	Juanillo Mc Andrew A.	Alcor @Planetary-hub

II. PROJECT INFORMATION

Project Title

An Error caused a bottleneck ◡^◡

Project Background

Traditional textbook assignments may not fully prepare computer science students for real-world problems, where small coding errors can cause major system failures. This project addresses this gap through a Java-based computer system and network simulation game that helps students develop practical debugging and problem-solving skills. In a low stakes environment.

Project Objectives

General Objective

To develop a Java-based application that will simulate a simplified version of real world systems pipeline and present it in the form of a game to make the learning and development of skill easier and more entertaining.

Specific Objectives

The project aims to:

- 1) To teach students how to handle complex system pipeline problems and develop practical logic and debugging skills through a gamified environment.
- 2) For educators to use as an interactive teaching application to demonstrate real-world programming and system architecture concepts.
- 3) For gamers and beginners to experience a fun, accessible entry point into software development without needing prior coding knowledge.



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Target Users

Target User	How They Will Use the System
Students	A platform where they can enhance their skills in coding by handling complex problems in a manner of a game.
Educators	Alternate learning environment for instructors to teach their students.
Gamers	Fun entertainment while being an entry point for people with zero experience in coding and up for challenge.

Proposed Features

Feature	Description
Simulated Node and Pipeline Management	Allows players to place, configure, and upgrade different network nodes on a grid to route simulated user traffic efficiently.
User Traffic and Load Simulation	Spawns varying types of in-game user objects with distinct request rates and patience levels, testing the pipeline's throughput and buffer limits under dynamic load spikes.
Artificial Bug Injection and Bottleneck Events	Periodically introduces systemic errors, latency spikes, and node failures that players must actively diagnose, debug, and resolve using UI controls to prevent total system crashes.
Stage Select and Level Progression System	Features multiple playable stages with increasing pipeline complexity, unique node constraint challenges, and customizable simulation speed settings.

III. JAVA AND OOP CONCEPTS

OOP Concept	How It Will Be Used
Classes and Objects	Objects such as nodes and users can be made as templates for use and for instantiation in other parts of the program.
Methods	Methods will be used to hasten the repetitive task to set up and instantiate the system nodes and user types from pre-made templates.
Encapsulation	All user actions between nodes and node functions and logic will be in each of their own classes to set up for easier inheritance.



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Inheritance	The many user types and system nodes have different behaviors, but they still share certain characteristics making it efficient to simply extend from existing parent classes.
Polymorphism	Given that each user type and system node have differing functions, functions in the parent class can be overridden by each individual user and node class for their own logic.
Abstraction	To account for the many user type classes and system node classes to extend from the node and user class, there can exist empty functions in both parent classes to account for the logic to its future children classes.
Exception Handling	For unexpected errors, it can be caught in code and revert the system state to a previous stable snapshot.
Arrays/Data Handling	The play field and the nodes having to contain many users and nodes, will have its data contained in one and two dimensional arrays.

IV. PROPOSED CLASSES

Class	Purpose	Main Attributes	Main Methods
PlayField	Holds the game grid, manages active game objects and updates the overall simulation state every tick.	grid: Node[][] uniUserList: User[] nodeMap: Node[] starRating: float	updateTick(): void renderGrid(): void getStarRating(): float checkStatus(): boolean
Node	Abstract base class for in-game node objects that receive and process users	nodeType: String capacity: int currentLoad: int userList: User[] inputRate: int outputRate: int status: String	processUser(user: User): User isOverloaded(): boolean updateNodeStatus(status): void



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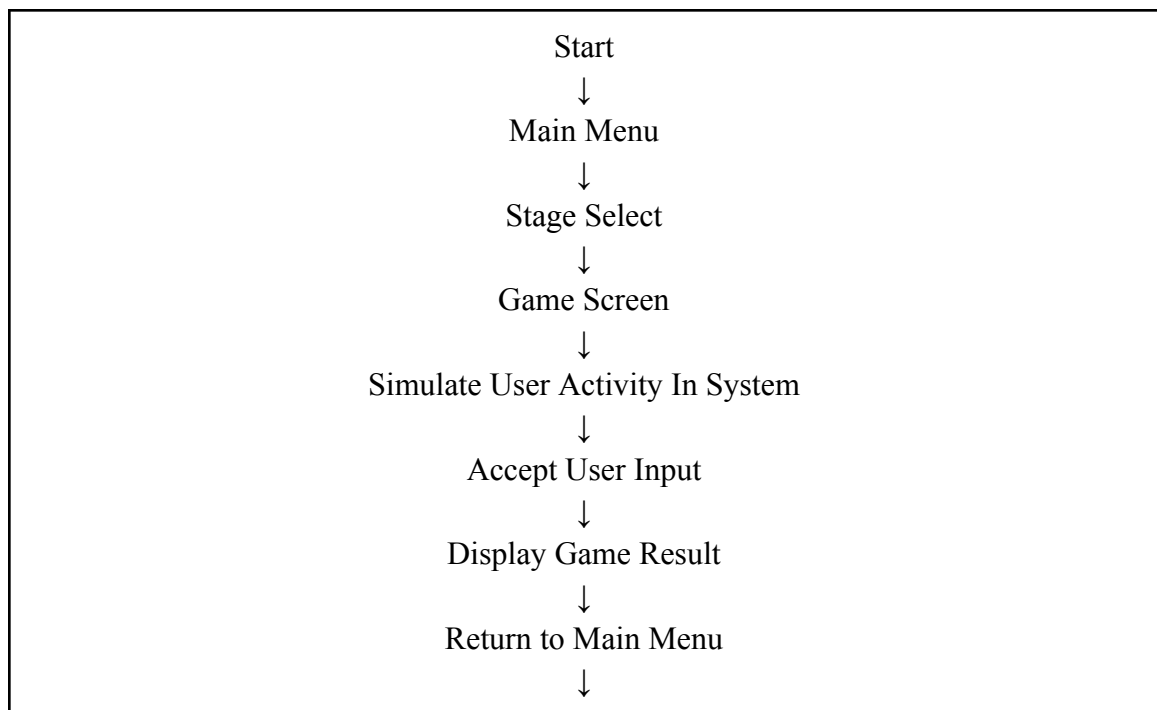
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User	Represents active in-game user objects moving through the system, generating data requests, and consuming node resources.	status: String rating: float userType: String	moveNextNode(currentNode: Node): void updateRating(delta: float): void getStatus(): String
Onboarding	In-game manager node responsible for spawning new User objects and introducing them into the pipeline.	inputRate: float outputRate: float	updateInputRate(rate: float): void updateOutputRate(rate: float): void
OffBoarding	In-game manager node responsible for catching existing user objects, clearing their data, and removing them from the system.	inputRate: float outputRate: float	updateInputRate(rate: float): void updateOutputRate(rate: float): void

The classes may be changed as the project is developed.

V. PROGRAM FLOW





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Exit

VI. TEAM RESPONSIBILITIES

Member	Assigned Tasks
Entienza, Ashenden Jairus	Programer, External Researcher
Hernandez, Diandre Daivik	Programer, Designer, Team-Lead
Juanillo, Mc Andrew	Programer, Designer

VII. DEVELOPMENT SCHEDULE

Week	Activity	Expected Output
Week 1	Planning and requirements	Project proposal
Week 2	System and class design	Class/UML diagram
Week 3	Initial coding	Basic classes
Week 4–5	Feature development	Working features
Week 6	OOP implementation	Completed OOP features
Week 7	Testing and debugging	Tested application
Week 8	Documentation and presentation	Final project

The schedule may be adjusted based on the course calendar.

VIII. EXPECTED OUTPUT

The game, simulating real world users in a computer network, will artificially raise simulated errors and bugs inside the simulated computer nodes that the user will try to solve in the manner of a game. The system will teach system management in such a way that players learn to analyze live data pipelines, identify network bottlenecks, and apply defensive logic to maintain optimal system health under pressure represented by a rating.



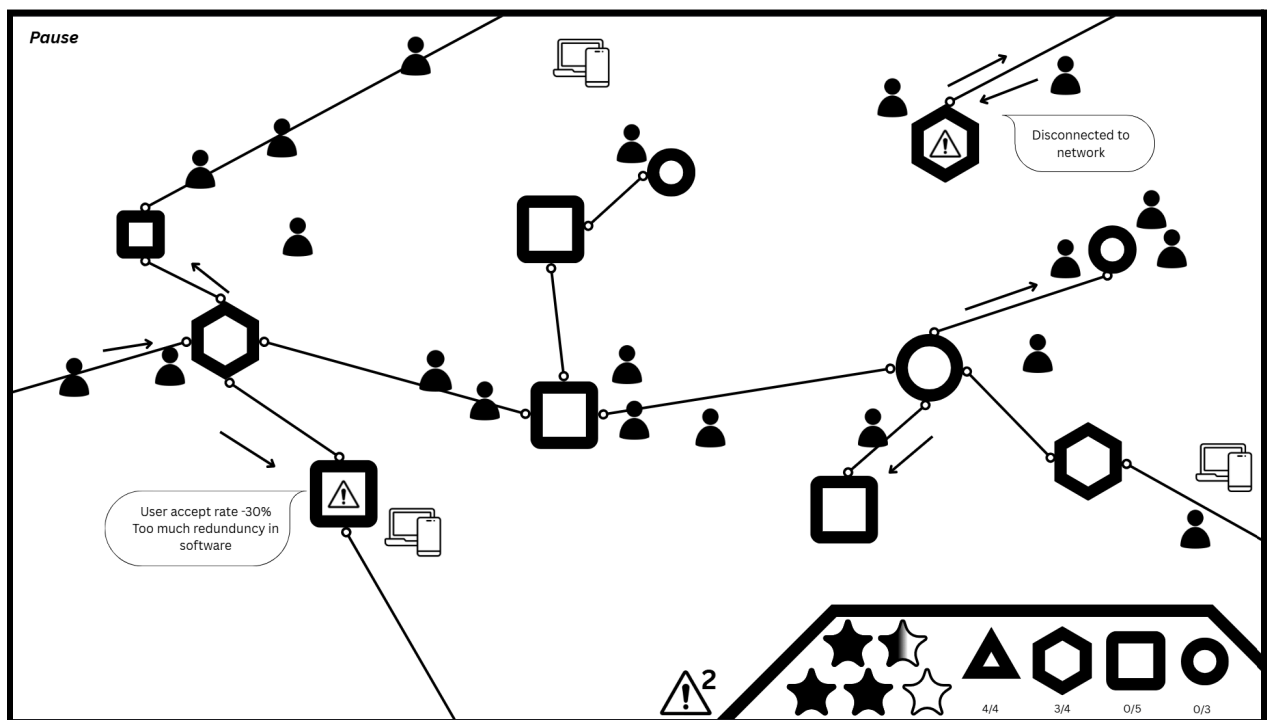
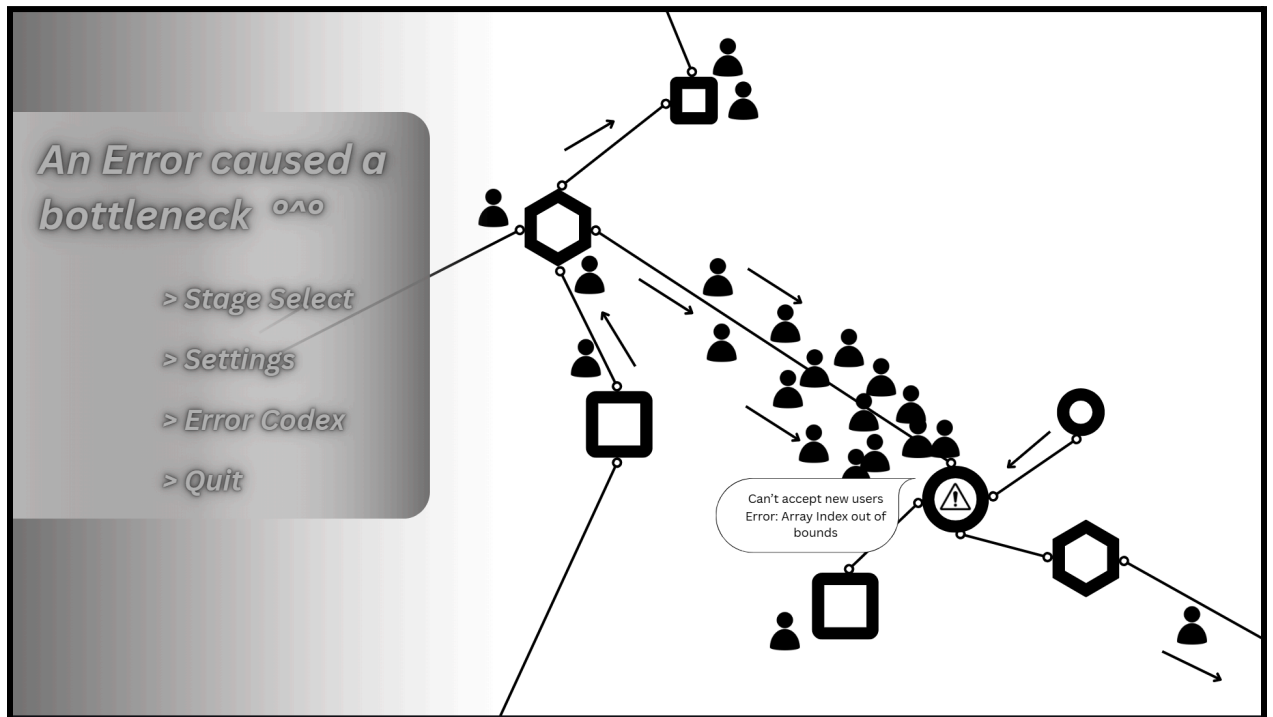
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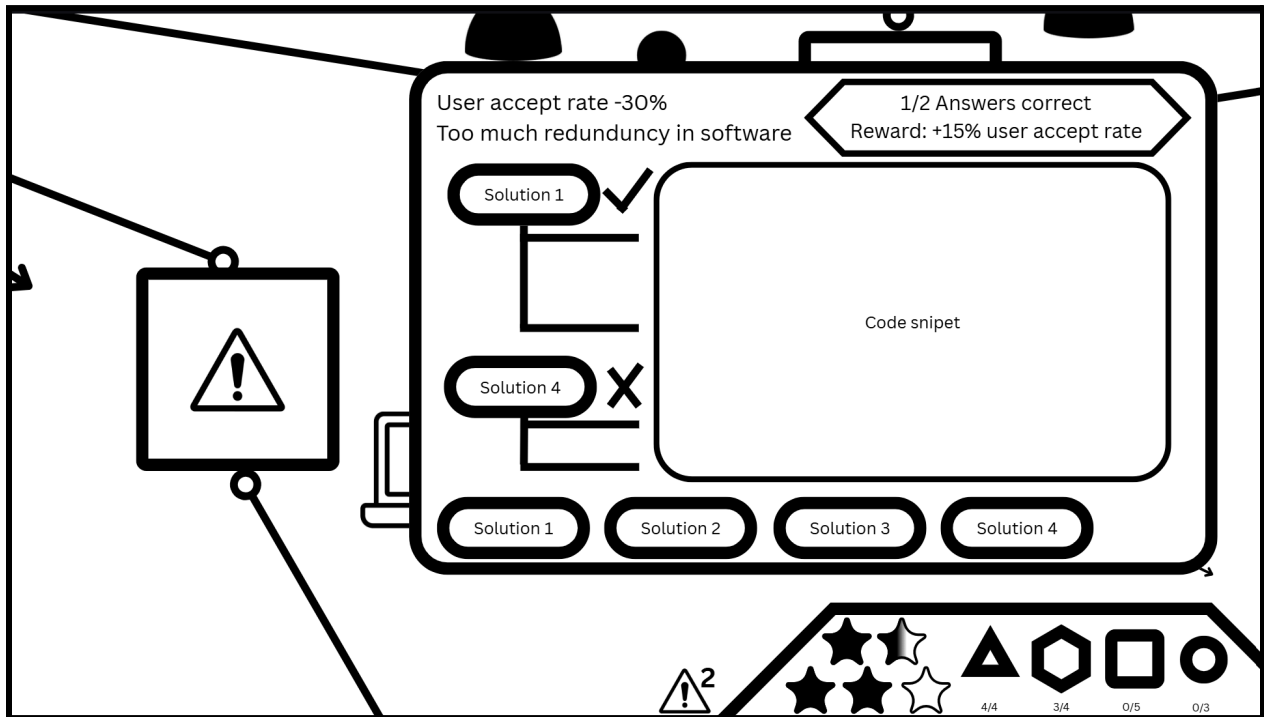


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IX. PROJECT LIMITATIONS

The project is strictly designed within the scope and timeframe of an introductory Object-Oriented Programming course. The application operates entirely as a local single-player desktop simulation, meaning it does not connect to real-world internet networks, send actual network packets, or interact with external servers or live protocols. Additionally, all progress tracking, stage states, and local settings are saved using standard local file handling rather than an external database system. The interface is built around standard 2D components and simple visual libraries rather than complex 3D game engines, and player interactions rely on strategic UI controls and configuration management rather than live code compilation or custom scripting within the game.

X. GITHUB REPOSITORY

Repository Name: [Placeholder]

Link to Repository: <https://github.com/Qlychn/Placeholder>

XI. APPROVAL

Prepared by:

Name	Signature
Entienza, Ashenden Jairus V.	
Hernandez, Diandre Daivik I.	
Juanillo Mc Andrew A.	



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Instructor's Comments:

Project Status:

- ☐ Approved
- ☐ Approved with Revisions
- ☐ Revise and Resubmit

Name of Instructor:

Ms. FATIMA MARIE P. AGDON, MSCS

Date Signed: _____